

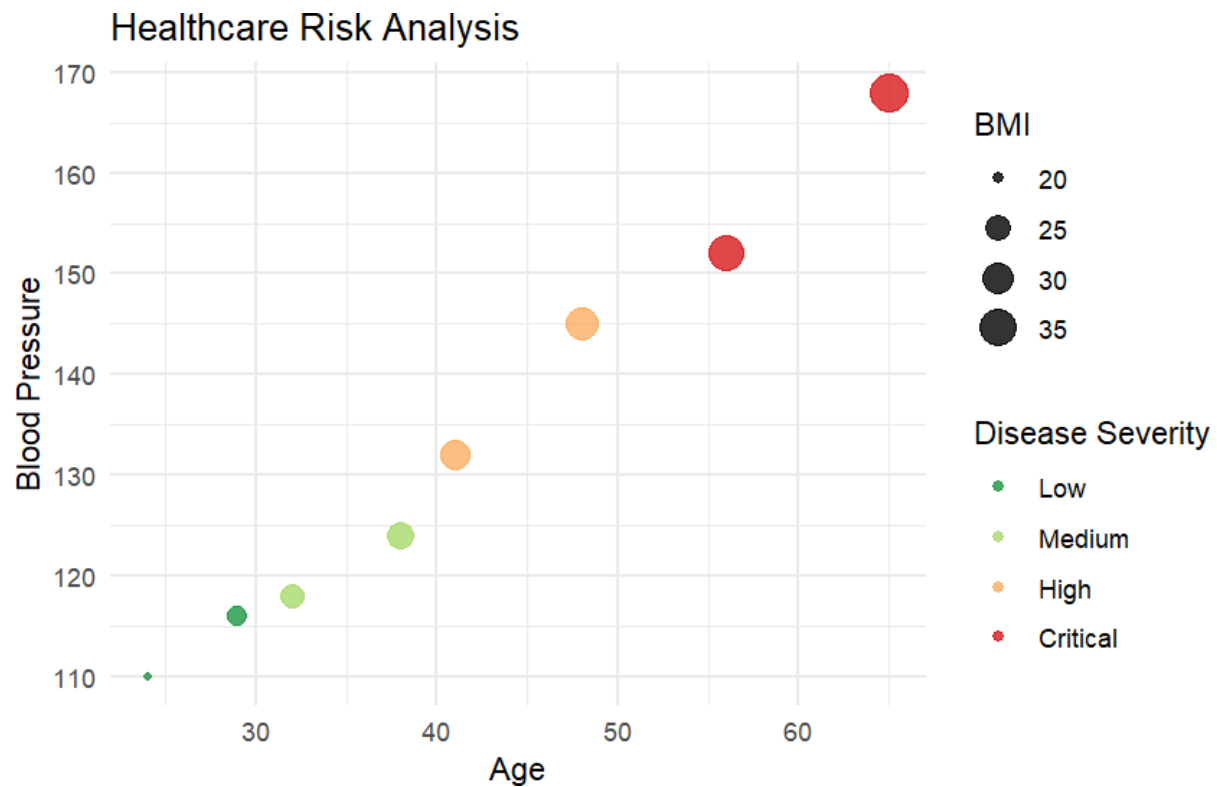
DSA0601 – DATA HANDLING

AND VISUALIZATION

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Question 1 – Healthcare Analytics



CODE:

```
# Install package if  
needed  
install.packages("ggplot2")
```

```
# Load library  
library(ggplot2)
```

```

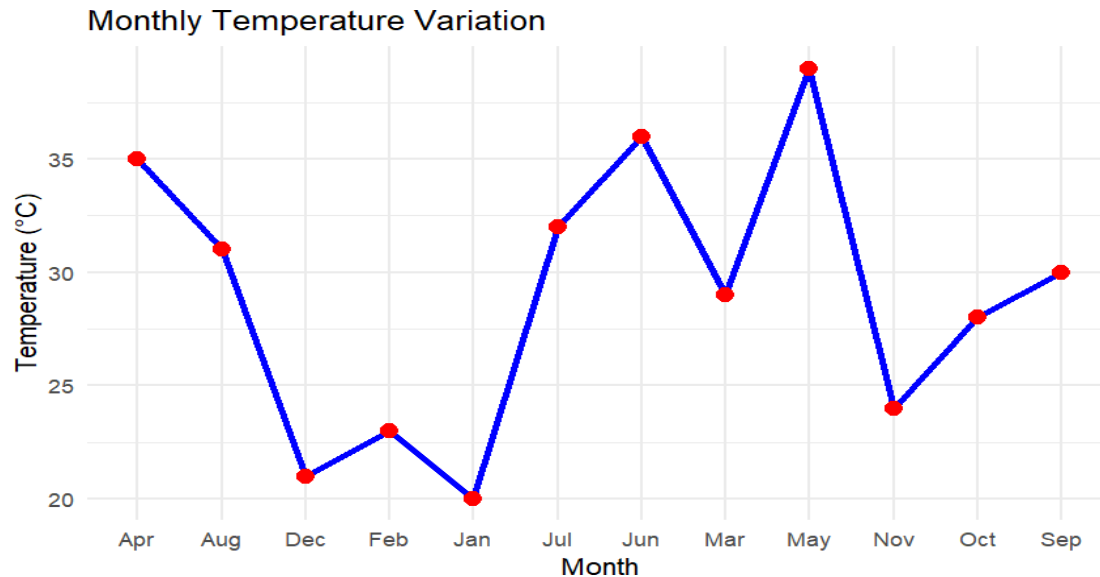
# Create dataset
health <-
data.frame(
  Patient=c("P1","P2","P3","P4","P5","P6","P7","P8"),
  Age=c(24,32,41,48,56,65,38,29),
  BMI=c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
  BloodPressure=c(110,118,132,145,152,168,124,116),
  Severity=factor(c("Low","Medium","High","High","Critical","Critical","Medium",
    "Low"), levels=c("Low","Medium","High","Critical"))
)
# Scatter Plot
ggplot(health,
  aes(x=Age,
    y=BloodPressur
    e,
    color=Severity,
    size=BMI)) +
  geom_point(alpha=0.8) +
  scale_color_brewer(palette="RdYlGn",
  direction=-1) + labs(title="Healthcare Risk
  Analysis",
  x="Age",
  y="Blood
  Pressure",
  color="Disease
  Severity",
  size="BMI") +
  theme_minimal()

```

Justification

- Age → X-axis: Shows how health risk changes with age.
- Blood Pressure → Y-axis: Indicates cardiovascular health.
- Disease Severity → Color: Green to red clearly represents increasing risk.
- BMI → Point Size: Larger points indicate higher BMI and greater obesity risk.
- The scatter plot effectively shows relationships among four medical variables simultaneously.

Question 2 – Climate Science



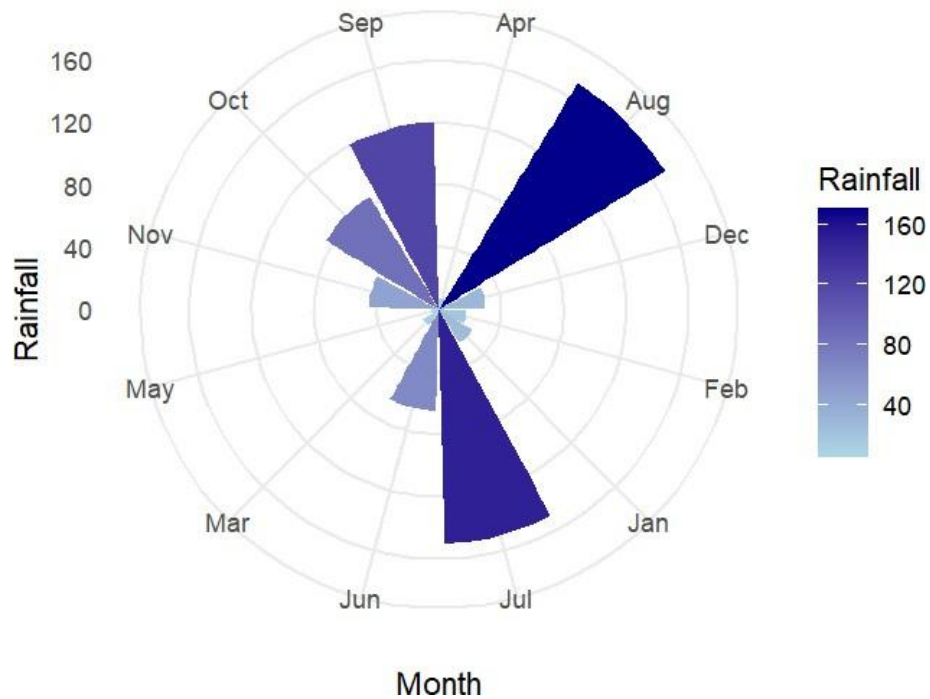
CODE:

```
library(ggplot2)

climate <- data.frame(
  Month=c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov",
    "Dec"), Solar=c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
  Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)

ggplot(climate,
  aes(x=Month,
    y=Temperature,
    group=1))+
  geom_line(color="blue",linewidth=1.2)
)+ geom_point(size=3,color="red")+
  labs(title="Monthly Temperature
    Variation",
    x="Month",
    y="Temperature
    (°C)")+
  theme_minimal()
```

Annual Rainfall Distribution



CODE:

```
ggplot(climate,
  aes(x=Month,
    y=Temperature,
    group=1))+
  geom_line(color="blue",linewidth=1.2)
)+ geom_point(size=3,color="red")+
  labs(title="Monthly Temperature
  Variation",
    x="Month",
    y="Temperature
    (°C)")+
  theme_minimal()
ggplot(climate,
  aes(x=Month,
    y=Rainfall,
```

```
    fill=Rainfall))+  
geom_bar(stat="identity")+  
coord_polar()+  
scale_fill_gradient(low="lightblue",  
                    high="darkblue")+  
labs(title="Annual Rainfall  
Distribution")+
```

theme_minimal()

Comparison

Cartesian

Chart

Advantages

- Easy to compare monthly temperature.
- Shows trends clearly.
- Suitable for continuous data.

Limitations

- Does not emphasize yearly cycle.

Polar Chart

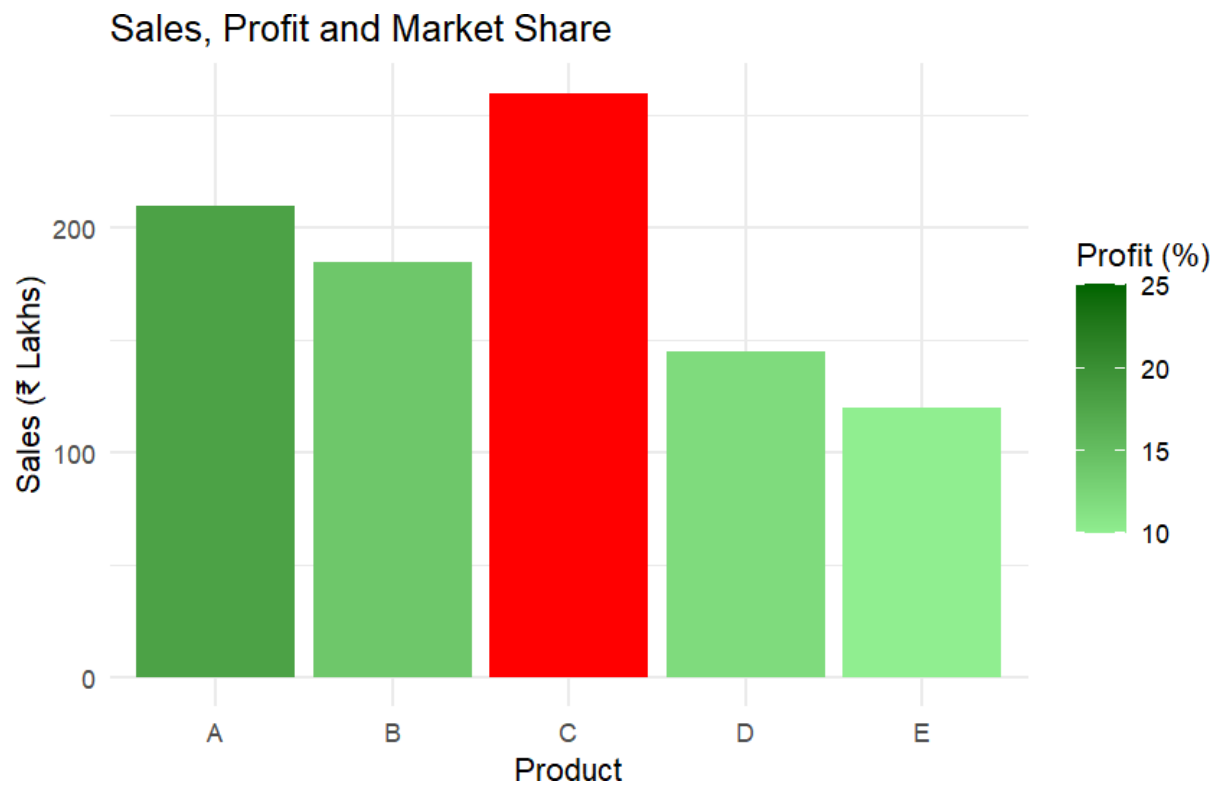
Advantages

- Represents seasonal rainfall patterns.
- Circular layout matches annual climate cycle.
- Visually attractive.

Limitations

- Exact values are harder to compare.
- Less accurate for quantitative comparison.

Question 3 – Business Analytics & Marketing



CODE:

```
library(ggplot2)
```

```
business <- data.frame(  
  Product=c("A","B","C","D","E"),  
  Sales=c(210,185,260,145,120),  
  Profit=c(18,14,25,12,10),  
  MarketShare=c(24,20,28,15,13)  
)
```

```
business$Highlight <- ifelse(business$MarketShare==  
  max(business$MarketShare),  
  "Highest","Others")
```

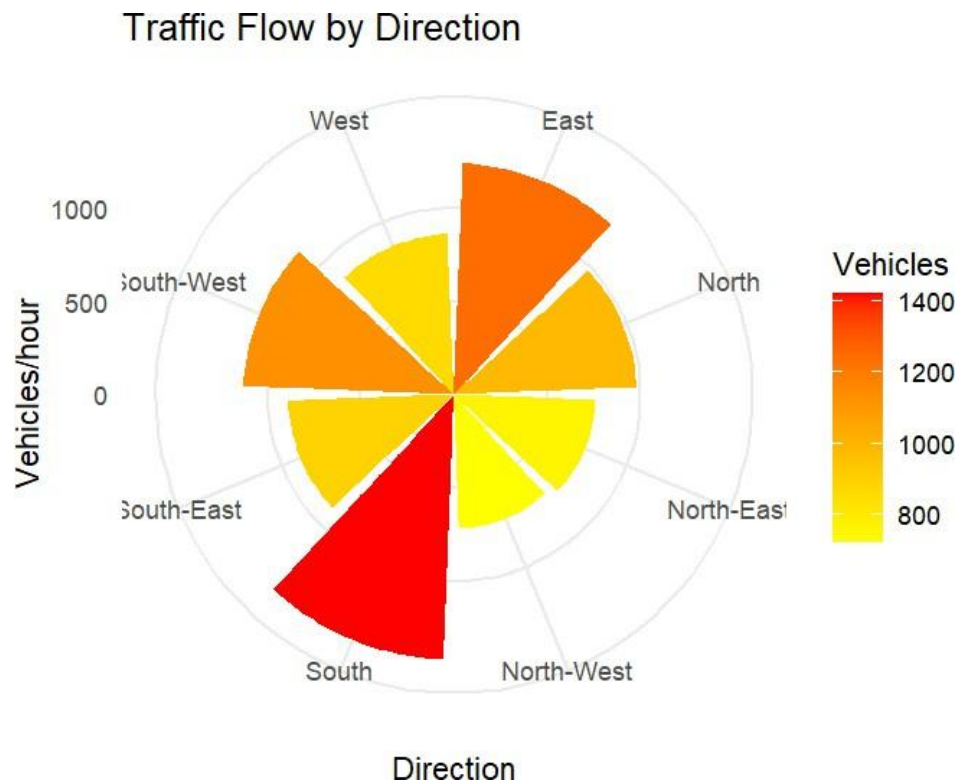
```
ggplot(business,  
  aes(x=Product,  
    y=Sales,  
    fill=Profit))+
```

```
geom_bar(stat="identity")+  
geom_bar(data=subset(business,Highlight=="Highest"),  
  fill="red",  
  stat="identity")  
+  
scale_fill_gradient(low="lightgreen",  
  high="darkgreen")+  
labs(title="Sales, Profit and Market  
Share",  
  x="Product",  
  y="Sales (₹  
Lakhs)",  
  fill="Profit  
(%)")+  
theme_minimal()
```

Justification

- **Bar Height** represents sales volume.
- **Sequential Green Palette** indicates increasing profit.
- **Red Highlight** immediately identifies the product with the highest market share.
- Helps managers quickly identify the best-performing product.

Question 4 – Transportation Engineering



CODE:

```
library(ggplot2)
traffic <-
data.frame(
  Direction=c("North","North-
    East","East","South-East",
    "South","South-
    West","West","North-West"),
  Vehicles=c(980,760,1240,890,1420,1130,860,720)
)
ggplot(traffic,
  aes(x=Direction,
    y=Vehicles,
    fill=Vehicles)) +
  geom_bar(stat="identity") +
  coord_polar() +
```

```

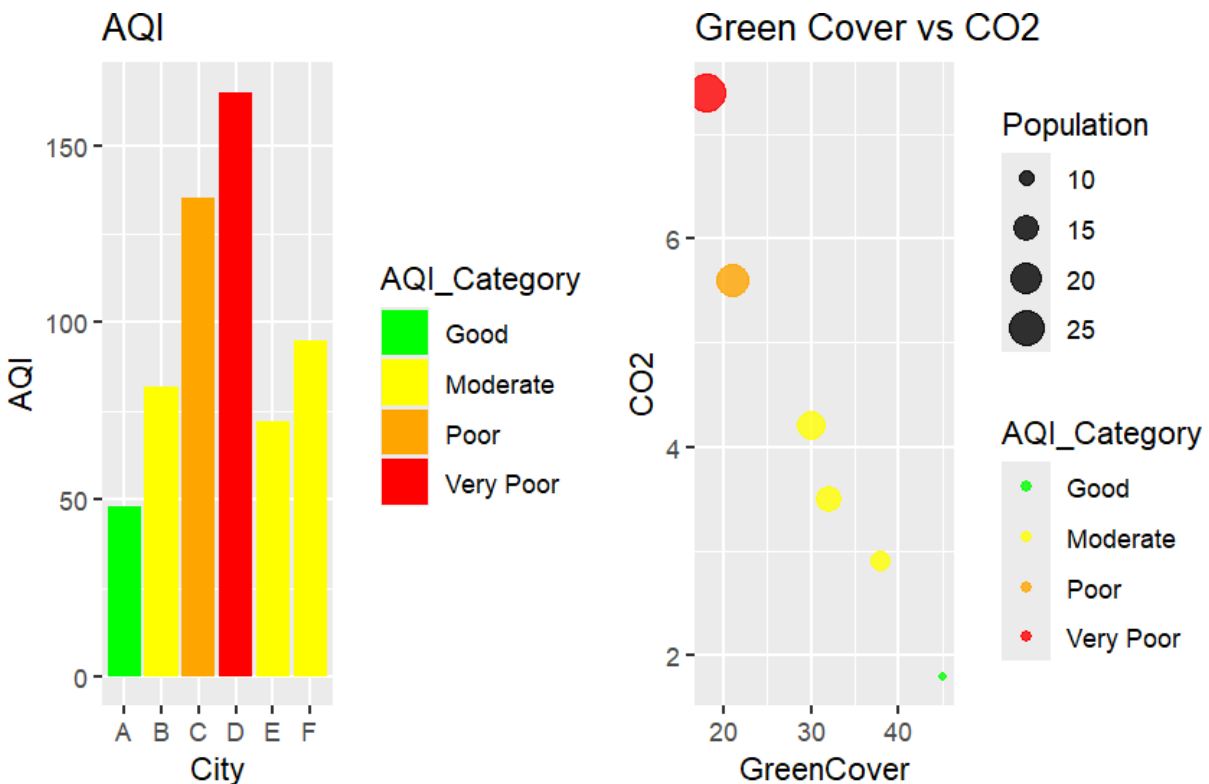
scale_fill_gradient(low="yellow
",
                    high="red")+
labs(title="Traffic Flow by
Direction",
      x="Direction",
      y="Vehicles/hour")
theme_minimal()

```

Justification

- Polar chart naturally represents compass directions.
- High-density traffic appears with darker colors.
- Easier to identify heavily congested directions.
- Better than Cartesian charts for directional data.

Question 5 – Environmental Sustainability Dashboard



CODE:

```
library(ggplot2)
library(gridExtra)
a) env <-
data.frame(
  City=c("A","B","C","D","E","F"),
  AQI=c(48,82,135,165,72,95),
  GreenCover=c(45,32,21,18,38,30),
  CO2=c(1.8,3.5,5.6,7.4,2.9,4.2),
  Population=c(9,15,22,28,12,18)
)
env$AQI_Category <-
  cut( env$AQI,
    breaks=c(0,50,100,150
    ,200),
    labels=c("Good","Moderate","Poor","Very Poor")
  )
# Plot 1
p1 <- ggplot(env,
  aes(x=City,
    y=
      y,
      y=AQI,
      fill=AQI_Category))+
  geom_bar(stat="identity")+
  scale_fill_manual(values=c("green",
    "yellow",
    "orange",
    "red"))
)+
  labs(title="AQI")
# Plot 2
p2 <- ggplot(env,
  aes(x=GreenCover,
    y=
      y=CO2,
      size=Population,
      color=AQI_Category))+
  geom_point(alpha=0.8)+
  scale_color_manual(values=c("green",
```

```
    "yellow",  
    "orange",  
    "red"))+  
labs(title="Green Cover vs  
CO2")  
grid.arrange(p1,p2,ncol=2)
```

Justification

- **Bar Chart:** Compares AQI across cities.
- **Scatter Plot:** Shows relationship between Green Cover and CO₂ emissions.
- **Point Size:** Represents population.
- **Color Palette:**
 - **Green → Good AQI**
 - **Yellow → Moderate**
 - **Orange → Poor**
 - **Red → Very Poor**
- The dashboard combines multiple environmental indicators into a single view, making it easier to identify pollution hotspots and support sustainability planning.